Pakhi Banchalia

Dresden, Germany | pakhibanchalia24@gmail.com | GitHub | Portfolio

Education

- **Technische Universität Dresden** *Oct 2025 - Present*
  Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
- **Amrita Vishwa Vidyapeetham Kerala, India** *2021-2025*
  Bachelor of Technology in Computer Science and Artificial Intelligence Amritapuri, India

Research Experience

- **Aalto University** *June 2026 - Present*
  Summer Research Intern Espoo, Finland
  - Conducting research on social intrinsic motivation and assistance in multi-agent reinforcement learning.
  - Studying information-theoretic approaches to agent autonomy, interaction, and supportive behavior.
- **Lossfunk** *July 2025 - Sept 2025*
  Research Intern Bangalore, India
  - Developed and evaluated homeostatic reinforcement-learning agents in the Crafter environment, using multiple internal variables such as hunger, energy, and safety to shape agent behaviour. [Code]
  - Designed experiments to study how alternative reward formulations and competing internal drives affect exploration, survival, behavioural diversity, and policy stability.
  - Compared homeostatic agents against standard reinforcement-learning baselines and analysed failure modes arising from drive imbalance, reward conflict, and changing environmental conditions.
- **Computational Machinery of Cognition (CMC) Lab, TU Dresden** *Nov 2024 - Present [Intermittent]*
  Research Intern (Independent Collaboration) Dresden, Germany
  - Designed RL agents with internal physiological drives to study retention and generalization across changing task distributions, focusing on the relationship between internal regulation and long-term behavioral stability.
  - Implemented a reward-modulated, gradient-free learning algorithm based on weight perturbations to investigate biologically plausible credit assignment without backpropagation.
- **Google Summer of Code, INCF** *May 2024 - Sept 2024*
  Contributor Remote
  - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization, representing developmental changes in cell state and connectivity over time.
  - Modeled evolving neural connectivity with Graph Neural Networks and used Topological Data Analysis with Kepler Mapper to inspect developmental and connectivity trajectories.
  - Delivered reproducible, documented open-source research code for computational developmental neuroscience. [Code]

Publications

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications*, ICDSA 2025. Lecture Notes in Networks and Systems, vol 1723. Springer, Cham. [DOI]

Projects

- **Aligned: Sponsorship Agent for TinyFish Accelerator**
  - Selected for the TinyFish Accelerator, a selective nine-week programme hosted by TinyFish and Mango Capital for founders building production-oriented agentic web applications.
  - Built an AI agent that discovers and evaluates relevant sponsorship opportunities by navigating live web sources and organizing potential sponsors for further review.
  - Developed a workflow for assessing sponsor relevance and supporting personalised outreach preparation. [Code]

- **SEANN: Self-Expanding and Adaptive Neural Networks**
  - Extended the CAN continual-learning framework with dynamic capacity expansion and task-relevant subnetworks.
  - Investigated structural plasticity as an alternative to fixed model scaling for reducing interference and catastrophic forgetting.
- **Dopamine: Reward-Modulated Gradient-Free Learning**
  - Implemented reward-modulated weight perturbation as an alternative credit-assignment mechanism and benchmarked early learning performance against gradient-based approaches.
  - Studied whether biologically inspired local updates can produce useful behavior without explicit backpropagation.

Talks and Posters

- **Brain-like features in homeostatic deep reinforcement learning agents @ Bernstein Conference 2025**
  - Poster presentation on homeostatic RL agents integrating internal and sensory inputs, showing structured behavioral and neural patterns
- **Emergence of Brain-like Features in Homeostatic RL Agents @ BrainBody 2025**
  - Oral presentation; demonstrated coupling between internal states, neural dynamics, and adaptive exploration

Technical Skills

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Agentic AI, Robotics, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, Weights & Biases (W&B)